

DAX MEASURES AND CALCULATED METRICS DOCUMENTATION

Introduction

DAX (Data Analysis Expressions) was used in Microsoft Power BI to create dynamic measures for analyzing the Sample Superstore dataset. The measures support the dashboard KPIs and allow business performance to be analyzed using filters and slicers such as Year, Region, Category, Customer Segment and Product.

The calculations focus on key areas including **sales, profitability, orders, customers, quantity sold, business growth and discount performance**.

Key DAX Measures

Total Sales calculates the total revenue generated:

Total Sales = SUM('Superstore'[Sales])

The total sales are approximately **\$2,297,200.86**. This is the main revenue KPI and can be analysed across different business dimensions.

Total Profit calculates the overall profit:

Total Profit = SUM('Superstore'[Profit])

The total profit is approximately **\$286,397.02**. It helps evaluate financial performance across products, customers, categories and regions.

Total Orders counts unique orders:

Total Orders = DISTINCTCOUNT('Superstore'[Order ID])

There are approximately **5,009 orders**. DISTINCTCOUNT is used because a single order can contain multiple transaction rows.

Total Customers counts unique customers:

Total Customers = DISTINCTCOUNT('Superstore'[Customer ID])

The dataset contains approximately **793 customers**, allowing customer-base and customer-performance analysis.

Profit Margin measures the percentage of sales retained as profit:

Profit Margin = DIVIDE([Total Profit], [Total Sales])

The overall profit margin is approximately **12.47%**. This provides a better indication of business efficiency than sales alone.

Average Sales per Order measures the average revenue generated per order:

Average Sales per Order = $\text{DIVIDE}([\text{Total Sales}], [\text{Total Orders}])$

The average is approximately **\$458.61 per order**.

Average Profit per Order measures the average profit generated per order:

Average Profit per Order = $\text{DIVIDE}([\text{Total Profit}], [\text{Total Orders}])$

The result is approximately **\$57.18 per order** and helps determine whether order volume is translating into meaningful profit.

Average Discount measures the average discount applied:

Average Discount = $\text{AVERAGE}(\text{'Superstore'}[Discount])$

The average discount is approximately **15.62%** and is useful for analysing the effect of discounting on profitability.

Time-Based and Business Performance Metrics

Sales Growth % measures the year-over-year change in sales using `CALCULATE`, `DATEADD` and `DIVIDE`. The analysis shows approximately **-2.83% growth in 2015, 29.47% in 2016 and 20.36% in 2017**. This indicates an initial decline followed by strong sales growth.

Profit Growth % uses a similar calculation to compare current profit with the previous year's profit. It helps determine whether profit is growing alongside sales.

Total Quantity Sold calculates the total number of units sold:

Total Quantity Sold = $\text{SUM}(\text{'Superstore'}[Quantity])$

This supports product and category volume analysis.

Average Profit per Customer calculates the average profit contribution per customer:

Average Profit per Customer = $\text{DIVIDE}([\text{Total Profit}], [\text{Total Customers}])$

This metric helps evaluate the overall profitability of the customer base.

Product and Discount Analysis

The **Loss-Making Products** measure identifies products with negative total profit. It uses FILTER and VALUES to count products that are generating losses. This helps management identify products that may require pricing changes, cost reductions or reduced discounting.

Profit Margin by Discount is calculated using:

Profit Margin by Discount = DIVIDE([Total Profit], [Total Sales])

When discount levels are placed on a visual, the measure dynamically calculates profitability for each level. The analysis shows that profitability decreases significantly as discounts increase:

- **0% discount:** 29.51% margin
- **20% discount:** 11.82% margin
- **30% discount:** -10.05% margin
- **40% discount:** -19.81% margin
- **50% discount:** -34.80% margin

This indicates that excessive discounting can lead to negative profitability and supports the recommendation that high discounts should be carefully controlled.

KPI Formatting

For clear presentation in Power BI, **Total Sales, Total Profit, Average Sales per Order and Average Profit per Order** should be formatted as currency. **Profit Margin, Average Discount, Sales Growth % and Profit Growth %** should be formatted as percentages. **Total Orders, Total Customers and Total Quantity Sold** should be formatted as whole numbers, with large values optionally displayed using abbreviations such as 5.01K.

Conclusion

The DAX measures transform raw Superstore transaction data into meaningful business information. They support the dashboard's KPIs and provide insights into **revenue, profitability, customers, orders, products, business growth and discounting**.

The analysis shows strong sales growth after 2015, while the discount analysis demonstrates that higher discounts are associated with declining and eventually negative profit margins. These findings provide useful support for management decisions involving **pricing, discount policies, product performance, customer value and overall profitability**.